X-band survey of galaxy cluster minihalos

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1. Background

Galaxy cluster minihalos – diffuse, steep-spectrum radio sources seen at the centres of relaxed galaxy clusters – are now known to be common, with $\approx 80\%$ of cool-core clusters hosting them (Giacintucci et al. 2017). However, it is still unclear how these electrons are accelerated to the relativistic energies required to produce the synchrotron emission. The emission extends on scales too large ($\approx$ 100–500 kpc) to be simply produced by relativistic electrons originating from the active galactic nucleus (AGN) of the central galaxy itself and travelling outward through the ICM via diffusion or jet action. One possibility ('re-acceleration') is that aged relativistic electrons from the AGN are re-accelerated to relativistic energies by turbulence (Gitti, Brunetti & Setti 2002) induced by sources including 'sloshing' of the ICM due to gravitational disturbances from e.g. minor mergers (Mazzotta & Giacintucci 2008). This mechanism seems supported by the observation that minihalo emission is often confined by ICM cold fronts detected in X-ray. An alternative possibility ('hadronic' or 'secondary' production) is that thermal protons throughout the cluster volume are accelerated to relativistic energies by supernovae, AGN or ICM shocks; these protons undergo interactions with the thermal proton population, producing pions which decay into a series of products including relativistic electrons (Pfrommer & Enßlin 2004).

In the 're-acceleration' scenario, the balance between re-acceleration and energy losses through synchrotron and inverse Compton radiation naturally creates high-frequency steepening of the radio spectrum, with the frequency at which this occurs dependent on the acceleration efficiency. Numerical simulations (ZuHone et al. 2013) of a cluster similar to the minihalo host RXJ1720.1+2638 have shown that an initial injection of relativistic electrons from an AGN, subsequently disturbed by sloshing, produces low-frequency spectra which have spectral indices ranging from $\alpha \approx 1.5 - 2.0$ (where $S \propto \nu^{-\alpha}$) and begin to steepen above ≈ 1 GHz.

In contrast, in the 'hadronic' scenario, the proton-proton interactions continuously refresh the population of relativistic electrons. Under relatively stable environmental conditions, a steady state is reached where the energy gains and losses balance out and the radio spectrum is a simple power-law, the spectral index of which depends only on the energy spectrum of the relativistic electrons. Similar simulations for this scenario (ZuHone et al. 2015; hereafter Z15) show that the turbulence generated by the merger does not disrupt this balance too much and the resulting integrated spectra follow a single power-law over a wide range in frequency with $\alpha \approx 1.1$. Fig. 1 (left) contrasts the simulated spectra produced by the two different production mechanisms. The authors note that although rapid magnetic field amplification (e.g. near cold front surfaces) has been shown to cause departure from steady-state and therefore spectral steepening, in their simulations they find the effect is barely observable when averaged over particles and line-of-sight, even at low observing frequencies where it should be maximized. *High-frequency* spectral steepening (as opposed to steepening of the *whole* spectrum at different *spatial locations*) therefore appears to only be a feature of the re-acceleration model.

Another contrasting feature of the 'hadronic' simulations from Z15 is the spatial extent of the minihalo. In this scenario, the emission is less tightly confined within the cold fronts than in the re-acceleration simulations for the reasonable choice of magnetic field initial structure and amplification used in their simulations. Z15 also note that Wiener et al. 2013 discuss momentum-dependent diffusion of the relativistic protons; although this effect is not included in the Z15 simulations they predict it would result in minihalos appearing broader at higher frequencies since higher-energy protons would travel further through the ICM. This prediction therefore provides another motivation for observing minihalos at *high radio frequency* to help to distinguish between the two production mechanisms and inform future numerical simulations.

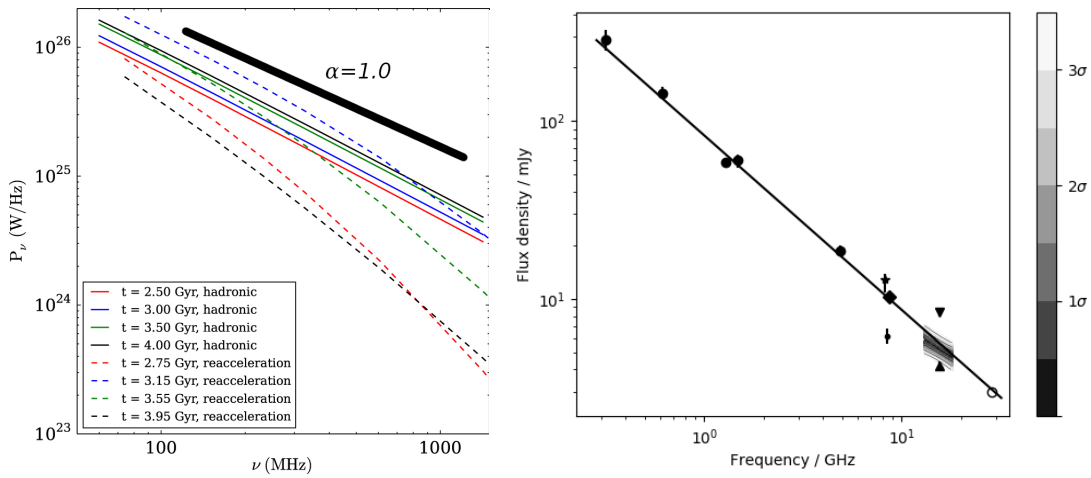

Figure 1: Left: integrated radio spectrum as predicted by numerical simulations at various times since the gravitational disturbance, for the 'hadronic' and 're-acceleration' models. The thick black line illustrates an unbroken $\alpha = 1.0$ power-law slope. Adapted from Z15. Right: observed integrated spectral energy distribution of the central part of the RXJ1720.1+2638 minihalo, from Perrott et al. 2021. The points marked with large circles and errorbars at < 5 GHz are taken from Giacintucci et al. 2014 (G14) and are used to fit the $\alpha = 1.0$ power-law spectrum shown with the solid line; the fit is extrapolated to higher frequency. The small circle with errorbar below the fit is the 8.4 GHz VLA measurement from G14. The star and diamond markers, displaced horizontally for clarity, show two reanalyses of the VLA 8.4 GHz data taking into account the *uv*-sampling. The $1 - 3\sigma$ constraints shown in greyscale are the results of *uv*-plane fitting to the AMI data; the triangles below and above are map-based measurements using the AMI Large Array (LA) and AMI-SA data respectively showing the effect of the different *uv*-sampling. The open circle is a 28.5 GHz measurement from Coble et al. 2007; it is an indication only as the SZ decrement has not been removed and the *uv*-sampling is not accounted for.

2. High-frequency observations of RXJ1720.1+2638

Giacintucci et al. 2014 (hereafter G14) analysed observations of the RXJ1720.1+2638 minihalo at a wide range of frequencies from 317 MHz to 8.4 GHz, finding that the integrated spectrum of the central part of the minihalo was consistent with a single power law with $\alpha = 1.0$ up to 4.86 GHz, with a tentative indication of steepening at 8.4 GHz; however the 8.4 GHz observations suffered from insufficient *uv*-coverage on the scale of the minihalo. We have recently analysed 13 – 18 GHz observations of RXJ1720.1+2638 with the Arcminute Microkelvin Imager (AMI; Zwart et al. 2008), using ancillary *Planck* and *Chandra* data to robustly remove the negative Sunyaev-Zel'dovich (SZ) effect signal (Perrott et al. 2021). As illustrated in the right-hand image in Fig. 1, these observations have shown that the spectrum continues with no steepening up to at least 18 GHz, and potentially up to 28.5 GHz based on a datapoint from Coble et al. 2007 (their table 1).

We also found that the surface brightness of the minihalo emission as seen at 15.5 GHz by AMI is not well fitted by the standard exponential model but requires a double-exponential model. The smaller-angular-scale model component is consistent with low-frequency observations and appears confined by the cold fronts, but the larger-angular-scale model component extends well past the cold fronts and does not appear to be detected in the lower-frequency observations of G14. This is not the only cluster where larger minihalo extent has been observed; Riseley et al. 2022 recently found that the minihalo in cluster MS 1455.0+2232, observed at 1.3 GHz with MeerKAT, extended further than previously thought and outside the X-ray cold fronts.

These two pieces of observational evidence fit remarkably well with the spectral and spatial factors discriminating between the 're-acceleration' and 'hadronic' production mechanisms according to the numerical simulations of Z15. The lack of observed spectral steepening could also be explained by continuous injection of electrons into the ICM by the AGN: we have recently observed the BCG at 0.05 arcsec resolution with e-MERLIN and shown the presence of sub-kpc jets (Perrott et al. 2023), and this would need to be incorporated into the numerical simulations to check if it is sufficient to explain the continuous spectrum. However, observing *both* a lack of spectral steepening *and* a spatial extent beyond the cold fronts seems to lend support to the 'hadronic' mechanism.

These observations raise several questions: is the RXJ1720 minihalo a special case, or is the lack of spectral steepening common amongst minihalos? Do all minihalos show larger spatial extent at higher frequency? These questions can be only be answered by a multi-frequency survey of cluster minihalos, with good sensitivity to large spatial scales as well as ~arcsec angular resolution to map out the spatial extent of the emission, rather than relying on a parametric model as required by the low resolution of the AMI observations.

3. Proposed observations

We propose to observe the sample of known minihalos between $-1^\circ < \delta < 30^\circ$ with the JVLA at X-band, in C-configuration. This sample has also been observed with LOFAR at 140 MHz and MeerKAT at 1.3 GHz (Riseley et al, data currently being processed). Combining the three datasets with similar angular resolutions will give a large spectral lever arm to test for the presence of high-frequency spectral steepening, and comparison of the spatial extents at the different frequencies will give insight into the question of high-frequency broadening. This will be the first time that a systematic study of minihalos has been made at high frequency: excluding the AMI result, a handful of individual minihalos have been detected at > 8 GHz (Sarazin, Baum and O'Dea 1995, Gitti, Feretti and Schindler 2006, G14) however these detections have been made pre-JVLA without the benefit of the JVLA bandwidth and sensitivity. Our proposed study is only possible with the JVLA due to the required frequency, angular resolution and sensitivity to extended scales.

The sample is listed in Table 1. We have used simulated observations to test the observability of the minihalos in the sample, extrapolating flux densities from the literature using a spectral index estimate if available, and otherwise conservatively assuming $S \propto \nu^{-1.5}$. We assumed an exponential model for the minihalo with an e-folding radius equal to a literature fit if available, and otherwise assumed $r_e = r/3$ where r is the literature overall radius estimate. We injected noise into the simulated observations consistent with the exposure calculator. To account for RFI, we decreased the bandwidth by 15% in the exposure calculator, and simulated the full bandwidth with the decreased sensitivity. An example simulated observation is shown in Fig. 2. We request observing time required per source of at least one hour (to obtain the uv-coverage required for mapping these extended sources), or the time required to obtain a 10σ detection of the simulated source with an appropriate uv-taper. This high-significance detection of the centre should allow us to also detect the lower-significance outskirts; we do not attempt to satisfy any more detailed significance requirements since the source structures are highly uncertain and the exponential model is representative only. For three sources, we find that due to the extent and/or faintness of the emission, D-configuration observations will be more suitable, and so we request a shorter 30 min observation of these sources to aid in compact source removal from D-configuration observations to be requested in the next D-configuration call. We exclude one source (A2029) from the sample since it is too extended even for D-configuration at these frequencies.

SZ contamination The minihalo emission will be contaminated by the SZ effect decrement, however at X-band the contamination is very small compared to the positive minihalo emission. We will subtract an SZ model based on ancillary *Planck* and X-ray data and do not expect this to be a major source of uncertainty.

Compact source contamination The fields will contain compact sources as well as the minihalos. Since the most significant minihalo emission will be on the shorter baselines, we will make use of a uniform weighting scheme and exclude shorter baselines to isolate and remove the emission from the compact sources as shown in Fig. 2 (right). We will also make use of ancillary data available at other frequencies, such as archival VLA data at lower frequency and/or higher frequency and higher resolution (not sensitive to the large scales of the minihalo) to help us to detect and remove sources.

Cluster	z	S_{10} / mJy	α	r_e / arcsec	Obs. time / hrs	Reference
A2626	0.0550	0.943	1.50	18.0	1	Gitti 2013
2A0335+096	0.0600	1.105	1.50	33.5	0.5*	Giacintucci et al. 2014b
A1795	0.0620	4.453	1.50	27.9	1	Giacintucci et al. 2014b
A2029	0.0765	1.021	1.50	61.7	excluded	Giacintucci et al. 2014b
A478	0.0880	0.870	1.50	32.4	0.5*	Giacintucci et al. 2014b
A1413	0.1430	0.090	1.52	12.2	0.5*	Trehaeven et al. 2023
A2204	0.1520	0.450	1.50	6.3	1	Giacintucci et al. 2014b
RXJ1720.1+2638	0.1644	8.314	1.10	16.9	1	G14
RXJ2129.6+0005	0.2350	0.279	1.20	15.0	4	Kale et al. 2015
MS1455.0+2232	0.2578	0.445	1.50	10.0	1	Giacintucci et al. 2014b
Z3146	0.2900	0.728	1.00	6.9	1	Giacintucci et al. 2014b
RXCJ1115.8+0129	0.3500	1.012	1.00	12.7	1	Trehaeven et al. 2023
ACT-CLJ0022.2-0036	0.8050	0.096	1.50	5.5	3	Knowles et al. 2019

Table 1: Sample of clusters to observe at X-band. 'S_{10}' is the predicted total flux density at 10 GHz assuming spectral index α. 'Obs. time' is the on-source X-band time requested. 'Reference' is the source of the lower-frequency flux density used for extrapolation and r_e. Sources marked with * will be more easily observed in D-configuration, so a short amount of time is requested here to aid in compact source removal.

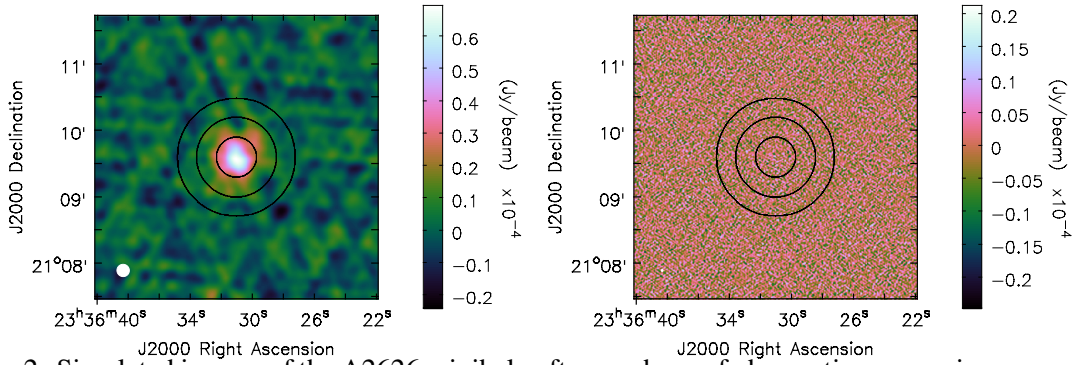

Figure 2: Simulated images of the A2626 minihalo after one hour of observation, assuming an exponential model with the parameters given in Table 1. To make the left hand image, a 10 kλ uv-taper has been applied resulting in an 11σ detection; the rms noise level is 6.1 μJy beam^{-1}. For the right hand image, uniform weighting has been applied which completely suppresses the resolved minihalo emission and will allow the detection and characterisation of compact sources; the rms noise level is 4.5 μJy beam^{-1}. On both images, the black contours show $[1, 2, 3] \times r_e$ (the e-folding radius).

Summary: total requested time (X-band): 15.5 hours on-source, 22 hours total including standard overheads.